

Influence of partial saturation on bulk modulus and attenuation: the experiment and the theoretical model

Lingyi He^{1,2}, Pinbo Ding^{1,2}, Bangrang Di^{1,2} and Feng Zhang^{1,2}

¹State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum (Beijing), Beijing 102249, China. E-mail: dingpinbo@cup.edu.cn

²CNPC Key Laboratory of Geophysical Exploration, China University of Petroleum (Beijing), Beijing 102249, China

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SUMMARY

Modelling the dispersion and attenuation of seismic waves in partially saturated rocks is important for quantitative interpretation of seismic data. To observe the partial saturation effects on modulus and attenuation, three sandstone samples with different porosities were used in the laboratory experiment. The samples were measured by ultrasonic testing system under different water saturation using drainage process. The experimental results show that the bulk modulus more rapidly increased at higher saturations (>80 per cent), and attenuation peaks could be observed. Then we develop a new partial saturation model based on Chapman's theory and poroelastic theory. The new model is consistent with the Gassmann equation when the water saturation is 0 or 100 per cent. By analysing the characteristic frequency, the new model can be concluded in the form of a standard linear solid model. A comparison of the new model with the White model and the measured data is conducted. The results show that the model performed well at predicting the effective bulk modulus and attenuation of the rock with different water saturation. Finally, we discussed the impact of frequency, fluids and pore structures on the new model. The results reveal that the new model will be helpful in discussing partial saturation in rocks.

Key words: Elasticity and anelasticity; Microstructure; Acoustic properties; Seismic attenuation; Wave propagation.

1 INTRODUCTION

The partial saturation of porous rocks is an important cause of attenuation and dispersion. Many geological settings are partially saturated in hydrocarbon reservoirs, such as the upper part of reservoirs containing gas, oil and water (Toms *et al.* 2006). The pore compressibility may vary when pores are saturated with different fluids, leading to pressure gradients between the connected pores, which result in a relative displacement between the fluid and solid. This relative motion then leads to dispersion and attenuation during wave propagation (Gist 1994). Determining the quantitative relationship between the fluid saturation and seismic attributes, such as wave velocity and attenuation, is a challenging problem in geophysics.

The effect of partial saturation on seismic wave propagation in rocks has been extensively studied Müller *et al.* (2010). Gregory (1976) showed that the results of *P*-wave velocity tests under partially saturated conditions differed from the predictions of the effective media theory based on the Wood model (Wood & Lindsay 1956). Subsequent experiments also suggested that the microscopic inhomogeneous fluid distribution causes this difference (Domenico 1976). These inhomogeneous fluid distribution are

related to drainage and imbibition process Cadoret *et al.* (1995). The drying process was found to lead to an inhomogeneous fluid distribution, whereas the depressurization process had a relatively homogeneous fluid distribution. Mavko (1994) showed that partial saturation under high-frequency leads to a nearly proportional increase in bulk modulus with saturation. The inhomogeneous distribution of fluids in pore rocks could be observed using CT imaging (Cadoret *et al.* 1995; Lebedev *et al.* 2009; Sun *et al.* 2022), and the discrepancies between the test results and theoretical models were highlighted. The ultrasonic results of Amalokwu *et al.* (2017) demonstrate that the relationship between bulk modulus and partial saturation is complex. And the explanations for the experimental data require multiple physical mechanisms.

Biot (1956a, b) proposed a theory of fluid saturated porous media, describing the energy dissipation of seismic wave fields owing to the relative motion between fluids and solids. However, according to the discrepancy between Biot's theory and experimental data (Batzie *et al.* 2001), researchers suggested that inhomogeneity of pore structure and fluid distribution at the micro- or mesoscopic scale has significant effects on seismic wave (Mavko & Nur 1975; Pride & Berryman 2003). The wave-induced fluid flow (WIFF) is caused by the fluid gradient is related to the differences in the pore

compressibility. White (1975) introduced the mesoscopic inhomogeneous fluid distribution, which described partially saturated rocks as concentric spheres with an outer layer of water and inner layer of gas. Dutta & Ode (1979) solved the White's model more rigorously by incorporating Biot's theory. Accordingly, the effect of partial saturation on the distribution of different fluids has been discussed extensively. Such as, the case of a random fluid distribution (Johnson 2001; Toms *et al.* 2006) and the case of a 1-D periodic distribution of fluids Müller & Gurevich (2004) were discussed. And the partial saturation model was further improved by incorporating additional factors like capillarity, interfacial impedance and crack (Kong *et al.* 2013; Qi *et al.* 2014a, b).

Chapman *et al.* (2002); Chapman (2003) discussed the effect of squirt-flow in fracture-pore systems owing to the inhomogeneity of the pore structure when rock is fully saturated. Papageorgiou & Chapman (2015) constructed a partial saturation model of fractured media by averaging fluid parameters (bulk modulus, viscosity, permeability, etc.). The dispersion and attenuation corresponding to the drainage and imbibition processes were also analysed. Later, the discussion of capillary forces were added to the studies of Papageorgiou *et al.* (2016); Papageorgiou & Chapman (2017). In these studies, the ratio of water-bearing pore pressure to gas-bearing pore pressure was assumed to be constant and a new partial saturation model was developed based on the previous study. Jin *et al.* (2018) discussed the effects of partial saturation on shear wave dispersion and anisotropy in cracked media. However, the Chapman's and Papageorgiou's models assume that the crack density is higher than 0. And only the fluid flow between the fractures and pores was included in the above effective media theories, whereas the fluid flow between the pores saturated with different fluids in the background medium was neglected.

In this study, we developed a partial saturation model in pore media based on the Chapman model. First, we performed ultrasonic tests on rocks of different porosities under partially saturated conditions. Subsequently, assuming a spherical pore structure, we established an equation of fluid motion between pores containing different fluids. Combined with the law of fluid mass conservation, the relationship between pore pressure and applied stress was solved under this condition. The partial saturation effective bulk modulus was solved using the interaction energy principle. The new model was extended to arbitrary pore structure by incorporating the quasi-static Biot theory, the theoretical prediction results were compared with the experimental data. Finally, numerical simulations of the new model were performed and the effects of frequency and fluids on the model were analysed and discussed.

2 EXPERIMENT METHODS

2.1 Sample preparation

Three synthetic sandstones were produced in this study using the cold-pressing method of Luan *et al.* (2016), which consisted of only quartz mixed with a small amount of cementing agent. Li *et al.* (2018) used SEM imaging to demonstrate that this method can produce well-sorted and homogeneous quartz sandstones. Ding *et al.* (2020) demonstrated that sandstones produced using this method have the same pore structure as natural sandstones and are more homogeneous. According to Vernik & Kachanov (2010), the rock frame can usually be discussed as a continuous medium when the porosity is smaller than the consolidated porosity (22–30 per cent). When the porosity is higher than the critical porosity (approximately 40 per cent), the contact model can be used to describe the

rock frame. The three quartz sandstones, with porosities of 4.97, 21.13 and 29.85 per cent (design porosities of 5, 22 and 30 per cent, respectively), which are consistent with the transition point proposed by Vernik & Kachanov (2010), represent low-, medium- and high-porosity sandstones, respectively.

During the sample construction, pure quartz sand was mixed with a small amount of epoxy resin adhesive. Then, the mixtures were placed in a specially designed mould and pressurized by a hydraulic press equipment for 100 hr at different pressures (Li *et al.* 2020). By applying different pressures during the production process, sandstone samples with different porosities can be obtained. The epoxy resin was the same as that used by Luan *et al.* (2016) and Li *et al.* (2020), YY-505 epoxy resin, which has an impact strength of 32–34, peel strength of 30–32, operability time of 2–3 hr and full cure time of 24 hr. This adhesive was demonstrated to be sufficient for synthetic sandstone construction. To minimize the interference of the epoxy resin with the results of the study, the percentage of epoxy resin in all samples was less than 5 per cent. Luan *et al.* (2016) suggested that a 5 per cent epoxy resin had no significant effect on the experiment. Therefore, it can be approximated that the modulus of mineral grains in the synthetic sample is the modulus of quartz.

2.2 Saturation method

The ultrasonic transmission test method was used in this experiment with a measuring frequency of 0.5 MHz. The experimental results reported by the previous literatures indicate that the method of sample saturation process causes different fluid distributions, where the drainage process yields an inhomogeneous distribution of the fluid (Cadoret *et al.* 1995; Ren S B 2020; Sun *et al.* 2022). Therefore, this method was used to measure samples with different saturation. The samples were first placed at 50 °C for approximately 48 hr to ensure that they were fully dried (Gregory 1976). Thereafter, the samples were placed in a vacuum reactor and saturated with degassed water. The samples in water were pressurized at 20 MPa for at least 24 hr. This process was continued until further pressurization no longer caused bubble effusion, and the sample weight remained constant within the test accuracy (0.001 g) to ensure complete saturation of the samples. Partial saturation experiments were performed at room temperature (approximately 20 °C), water saturation decreased until the mass did not decrease with increasing time. Velocity and attenuation ultrasonic tests were performed during the drying process. Before each test, the mass of the samples was measured and used to calculate the saturation. Finally, the sample was redried at 50 °C for approximately 48 hr to measure the elastic wave velocity and attenuation after fully dries.

2.3 Velocity and attenuation estimation

We use the ultrasonic pulse transmission method to measure elastic wave velocities. The experimental set-up and the testing process are shown in Fig. 1. We obtained the total travel time using the method shown in Fig. 1(c). The transit time of the wave in the sample can be obtained after zero correction. Then the rock velocities are simply determined by the ratio of sample length and the transit time.

Attenuation was measured using the commonly used spectral amplitude ratio method (Toksöz *et al.* 1979).

$$\ln \left(\frac{A_1}{A_2} \right) = x (\alpha_1 - \alpha_2), \quad (1)$$

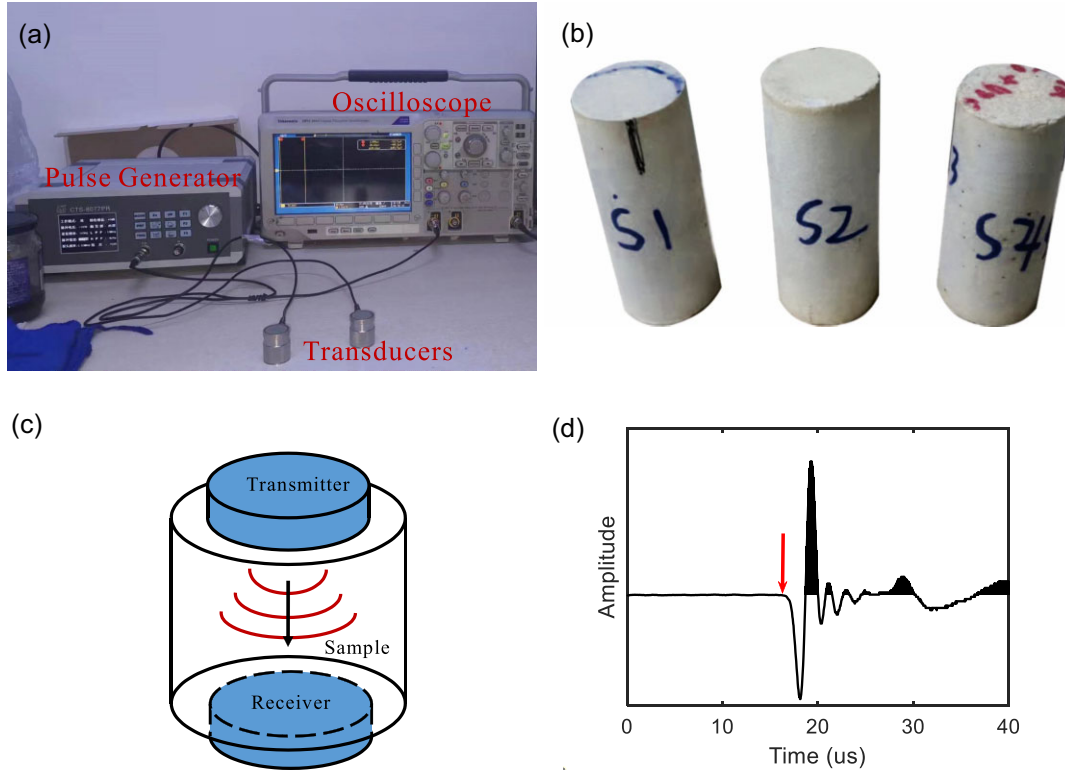


Figure 1. (a) The experiment set-up, (b) the samples used in the experiments, (c) the schematic of ultrasonic pulse transmission measurement and (d) the method of picking the first break.

where x is the sample length; A_1 and A_2 represent the received signal amplitude spectrum of the reference and test samples, respectively; α_1 and α_2 represent the attenuation coefficient of the reference sample and test sample, respectively; and the attenuation coefficient has the following relationship with the quality factor Q .

$$\alpha \approx \frac{\pi f}{Qv}, \quad (2)$$

where f is the frequency and v is the velocity. Here, an aluminium sample was used as a reference sample, which had a P-wave velocity of 6400 m/s. The aluminium block had a very large quality factor (approximately 150 000 Zemanek & Rudnick 1961); thus, its attenuation can be approximated as 0 to obtain the quality factor of the sample to be measured using eq. (1). The results of elastic wave velocity and Q in this experiment are presented in Tables A1–A3.

3 EXPERIMENT RESULTS

Fig. 2 shows the variation in the velocities and bulk modulus with saturation for samples with different porosities. In Figs 2(a)–(c) as the water saturation increases, the P -wave velocity first decreases slightly before increasing. The P -wave velocity of saturated samples is greater than that of dry samples, and this difference increases with the porosity. Meanwhile, with the increase in saturation, the S -wave velocity decreases. This is owing to that the increase in effective density of the sample has more substantial effect than the influence of the fluid on the shear modulus. The samples with high porosity has a large difference between fully water-saturated and dried S -wave velocity. The bulk moduli of the three samples increased with increasing water saturation. The larger the porosity of the samples, the greater the increase in the saturation bulk modulus

relative to the dry bulk modulus. The bulk modulus of the sample S2 (29.85 per cent) grew slowly at low-saturation (<80 per cent), then increase significantly as the water saturation further increased until the sample was completely saturated. The bulk modulus of S1 (21.03 per cent) also increased faster at higher saturations. However, the threshold of change was at a lower saturation of approximately 70 per cent. The transition point was much lower at approximately 30 per cent in the test results for S4 (4.97 per cent), and then the increase in the bulk modulus was approximately proportional to the water saturation. this is consistent with the previous results in the study reported by Si (2016).

Fig. 3 shows the variation in the logarithmic amplitude ratios $\ln(A_1/A_2)$ and attenuations of P wave with saturation for samples at different porosities. We rearrange the curves in Fig. 3 to see the data clearly by adding or subtracting a constant to the spectral ratio. And it won't change the measured value of Q . Figs 3(a)–(c) show that the spectral ratio exhibits good linearity with frequency. But in Fig. 3(c), the testing results with low amplitudes and strong attenuation are disturbed by noise, but still maintain a certain linearity. Error bars indicate the standard deviation of the measured $1/Q_P$. The average error of Q_P in this experiment was approximately 13.8 per cent. The results of this measurement is relatively reliable comparing to the similar laboratory studies, such as the error of Q_P measurement is 15 per cent as reported by Johnston & Toksöz (1980). The P -wave attenuation for the samples with different porosities increased with water saturation. The P -wave attenuation increased almost linearly with saturation at low saturation, consisting with the partially saturated measurements reported by Mavko (1994). The attenuation reached a maximum at saturations close to 90 per cent for the samples with 21.03 and 29.85 per cent porosity, whereas the strongest attenuation of the 21.03 per cent sample was at a smaller

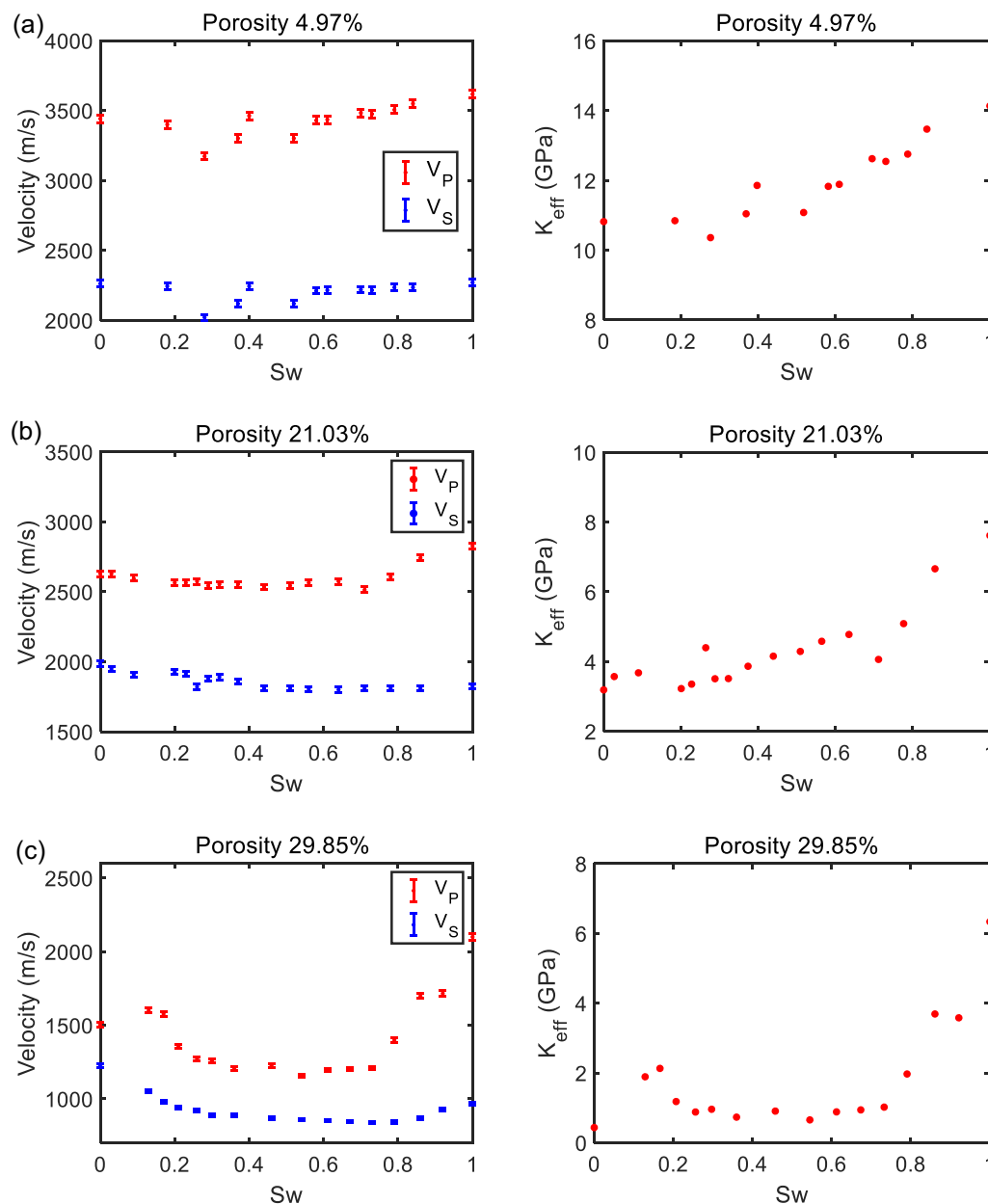


Figure 2. Water saturation dependence of velocities V_P and V_S and bulk modulus K_{eff} measured for samples with different porosities.

saturation (closer to 80 per cent). In contrast, this peak attenuation was not clearly observed in the test results for 4.97 per cent of the samples.

4 THEORY

As the experimental results show, the saturation has great impact on the elastic modulus and attenuation when water and air present in the rock. Therefore, we aim to theoretically illustrate the effect of fluids in partially saturated samples. A squirt flow theory usually aims at the fluid-saturated media (e.g. Dvorkin *et al.* 1995; Chapman 2009). However, local flow is induced by fluid pressure gradients. With the calculation of the fluid pressure gradient, we can describe the fluid flow within the pores, regardless of whether the pressure gradient is caused by the fluid distribution or the pore structure. So, we can model partial saturation using the following steps similar

to those of squirt flow model: (1) developing an equation to relate the fluid pressure and applied stress; (2) proposing an equation to describe the fluid exchange between WP and GP due to differences in fluid pressure and (3) calculating the effective elastic modulus due to the fluid flow.

4.1 Partial saturation model with spherical pores

The Chapman model and its extensions are based on the inclusion theory (Eshelby 1957). As previously mentioned, the Chapman model and its extensions only consider the squirt-flow of the same fluid between the crack-pore structures when establishing the fluid equation of motion, thereby neglecting the pressure difference in the background pores containing different fluids. Although Papageorgiou & Chapman (2017) extended the Chapman model to partial saturation conditions. The assumptions still limit the model to crack

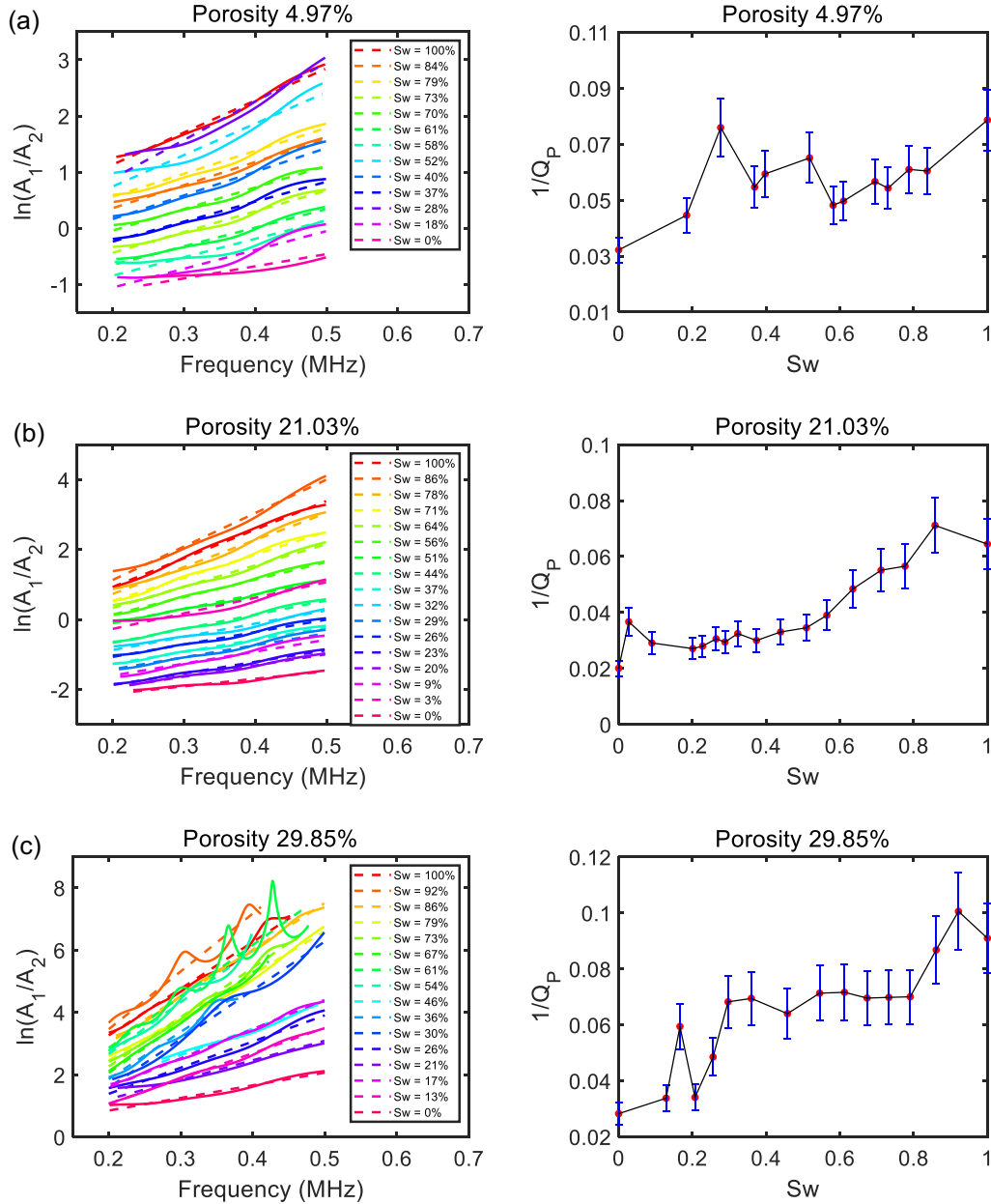


Figure 3. Water saturation dependence of the logarithmic amplitude ratios $\ln(A_1/A_2)$ and attenuations $1/Q_p$ measured for samples with different porosities. The solid lines in figures about $\ln(A_1/A_2)$ are the measured results and the dashed lines are the linear fitting curves.

densities higher than 0. Therefore, we first discussed the effect of the fluid flow when the spherical pores contained different fluids based on Chapman's theory. The schematic of the system is shown in Fig. 4.

The pores were assumed to be fully connected. During the wave propagation, the fluid flow between the two types of pores is caused by the different compressibility of the two types of fluids. Consider the average mass of the fluid in a certain pore is m . The fluid flow between pores containing different fluids can be expressed using Darcy's law.

$$\partial_t m_i = \frac{\rho \kappa \zeta}{\eta} (p_j - p_i), \quad (3)$$

where $i \neq j = 1, 2$, referring to water-bearing pores (WP) and gas-bearing pores (GP), respectively. And ρ is the density of the fluid

flowing between the pores, κ and η are the permeability and viscosity of the fluid, respectively and ζ is the length scale controlling the fluid flow.

Chapman *et al.* (2002) described the relationship between applied stress σ , pore pressure p and pore volume change in spherical pores as follows:

$$V_p = V_p^0 \left(1 - \frac{3}{4\mu} \frac{1-\nu}{1+\nu} \sigma_{II} + \frac{3}{4\mu} p \right), \quad (4)$$

where V_p is the pore volume, V_p^0 is the initial pore volume and μ and ν are the shear modulus and Poisson ratio of the solid background, respectively. The relationship between density and fluid pressure is represented by the following:

$$\rho = \frac{1}{1 - \frac{p}{K_f}} \simeq \rho_0 \left(1 + \frac{p}{K_f} \right), \quad (5)$$

Squirt-flow in the spherical pore system

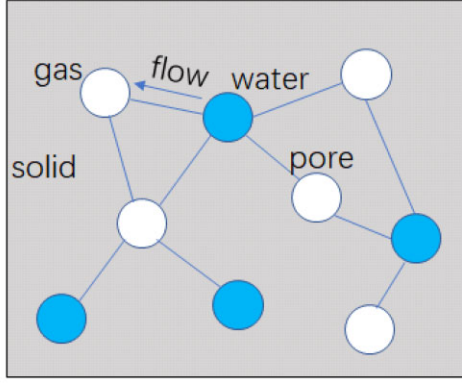


Figure 4. Schematic diagram of the partial saturation model assuming the pores are spherical.

where K_f is the bulk modulus of the fluid and ρ_0 is the undisturbed density of the fluid. The volume strain and mass change were significantly lower than 1. Substituting eq. (5) into eq. (4) and ignoring the higher-order terms, the mass change of the spherical inclusions can be obtained as follows:

$$dm = \frac{3m^0}{4\mu} \left[(1 + K_p) p - \frac{1 - \nu}{1 + \nu} \sigma_{ll} \right], \quad (6)$$

where $K_p = \frac{4\mu}{3K_f}$, $m_0 = \frac{4}{3}\pi a^3 \rho_0$. When the WP is connected to the GP, the mass of water in the WP decreases and water flows into the GP. Subsequently, the increase in fluid mass in the GP is determined by the injection of water. Therefore, according to the above conditions, the following is obtained:

$$\begin{aligned} \partial_t m_1 &= \frac{3m^0}{4\mu} \left[(1 + K_{p1}) \partial_t p_1 - \frac{1 - \nu}{1 + \nu} \partial_t \sigma_{ll} \right] \\ \partial_t m_2 &= \frac{3m^0}{4\mu} \left[(1 + K_{p2}) \partial_t p_2 - \frac{1 - \nu}{1 + \nu} \partial_t \sigma_{ll} \right]. \end{aligned} \quad (7)$$

Combining eq. (3), the relationship between the pressure in the two pores and the applied stress is obtained. To simplify this expression, we set

$$\begin{aligned} \tau_p &= \frac{\pi l (1 + K_{p1})}{\mu} \left(\frac{\eta}{\kappa} \right) a \\ \gamma &= \frac{1}{(1 + K_{p1})} \frac{1 - \nu}{1 + \nu} \end{aligned} \quad (8)$$

where a and l are the pore radius and squirt length, respectively. To solve the dispersion of the elastic modulus, by transforming the above equation into the frequency domain, the following is obtained:

$$\tilde{p}_1 = \frac{1}{1 + i\omega\tau_p} (\tilde{p}_2 + i\omega\gamma\tau_p\sigma_{ll}). \quad (9)$$

We can note that the pressure of fluid 1 is not constantly proportional to pressure of fluid 2. And the relationship between p_1 and p_2 varies with parameters, such as applied stress, grain modulus and frequency, which is different from the assumption of Papageorgiou & Chapman (2017). From the conservation of pore fluid mass, the following is obtained:

$$\partial_t m_1 + \partial_t m_2 = 0. \quad (10)$$

The mass here is the total mass of the fluid, hence the effects of porosity and saturation must be considered in the calculation. Define

two intermediate variables:

$$\begin{aligned} S(\omega) &= \frac{i\omega\gamma\tau_p}{1 + i\omega\tau_p} \\ F(\omega) &= \frac{1}{1 + i\omega\tau_p}. \end{aligned} \quad (11)$$

The following can be obtained:

$$\tilde{p}_2 = H_{ij}^2(\omega) \tilde{\sigma}_{ij}, \quad (12)$$

where

$$H_{ij}^2(\omega) = \frac{\frac{1-\nu}{1+\nu} - S_w (1 + K_{p1}) S(\omega)}{S_w F(\omega) (1 + K_{p1}) + (1 - S_w) (1 + K_{p2})} \delta_{ij}. \quad (13)$$

According to eq. (9), the following can be obtained

$$\begin{aligned} \tilde{p}_1 &= H_{ij}^1(\omega) \tilde{\sigma}_{ij} \\ H_{ij}^1(\omega) &= F(\omega) H_{ij}^2(\omega) + S(\omega) \delta_{ij}. \end{aligned} \quad (14)$$

Here, the relationship between the two fluid pressures and the applied stress is obtained. According to Eshelby (1957), the interaction energy principle is as follows:

$$C_{ijkl}^{eff} \varepsilon_{ij} \varepsilon_{kl} = C_{ijkl}^0 \varepsilon_{ij} \varepsilon_{kl} - \sum_n \phi_n (\varepsilon_{ij}^{inc} \sigma_{ij} - \sigma_{ij}^{inc} \varepsilon_{ij}). \quad (15)$$

To calculate the effective bulk modulus, the following is established

$$\begin{aligned} \varepsilon_{ij} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \sigma_{ij} &= \sigma \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (16)$$

For spherical pores, the expression of the applied stress σ can be found in Chapman (2009), and the expression of the effective bulk modulus can be obtained.

$$K_{eff} = K_m - \frac{K_m^2}{\sigma^2} \sum_n \phi_n (\varepsilon_{ij}^{inc} \sigma_{ij} - \varepsilon_{ij} \sigma_{ij}^{inc}), \quad (17)$$

where K_{eff} and K_m are the effective bulk modulus and grain bulk modulus, respectively. The stress and strain of the inclusions are expressed as follows:

$$\begin{aligned} \sigma_{ij}^{inc} &= \begin{bmatrix} \tilde{p} & 0 & 0 \\ 0 & \tilde{p} & 0 \\ 0 & 0 & \tilde{p} \end{bmatrix} \\ \theta^{inc} = \varepsilon_{ll}^{inc} &= \frac{3}{4\mu} \frac{1 - \nu}{1 + \nu} \sigma_{ll} - \frac{3}{4\mu} \tilde{p}. \end{aligned} \quad (18)$$

Therefore, the expression for the effective bulk modulus as a result is obtained as follows:

$$\begin{aligned} K_{eff} &= K_{dry} + \phi \frac{K_m^2}{\sigma} \left(\frac{3}{4\mu} p^* + \frac{1}{K_m} p^* \right) \\ K_{dry} &= K_m - K_m^2 \phi \frac{9}{4\mu} \frac{1 - \nu}{1 + \nu}, \end{aligned} \quad (19)$$

where the effective pore pressure can be approximated using volume averaging.

$$\phi p^* = \phi_1 \tilde{p}_1 + \phi_2 \tilde{p}_2, \quad (20)$$

where $\phi_1 = \phi S_w$ and $\phi_2 = (1 - S_w)\phi$ denote the volume fractions of the pores of the two fluids. The shear modulus is considered as fluid

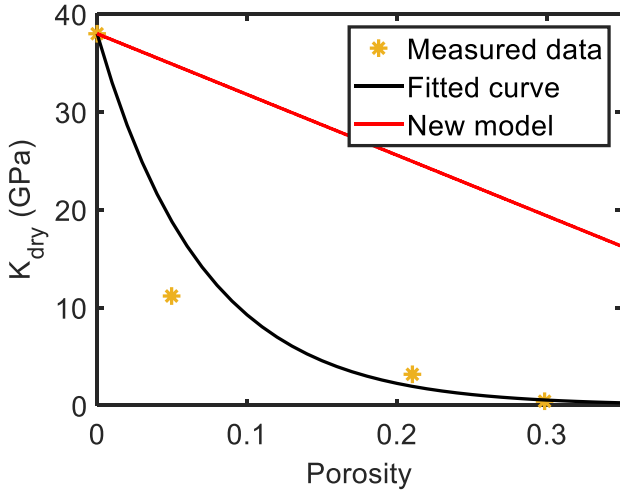


Figure 5. Comparison of the measured dry bulk modulus and the prediction.

independent. And we set it has the same expression as in Chapman *et al.* (2002). Therefore, the shear modulus is:

$$\mu_{\text{eff}} = \mu - 15\phi\mu \frac{1-\nu}{7-5\nu}. \quad (21)$$

Comparing to Chapman (2003) which containing both the pores and fractures, eq. (21) only consider the porous media without the fractures. Based on eq. (19), K_{dry} tends to decrease linearly with porosity when p^* is equal to 0 (dry condition). The reason for this result is that the Chapman model is based on Eshelby's theory, which assumes that a single pore is far away from the applied stress. To satisfy this assumption, the pore needs to be isolated. Such assumption leads to a significant overestimation of the dry modulus of rock when describing the pore medium. When fluid-induced dispersion and attenuation are discussed, overestimation of the dry bulk modulus leads to a low attenuation predicted by many theories (e.g. White's model and Biot's theory). The Fig. 5 suggests that an assumption more general than spherical pores is needed to illustrate the experimental data.

4.2 A generalized model

Referencing the inclusion theory, a generalized model without restricting the pore shape was derived using the poroelastic theory. First, the equation of fluid motion is the same as eq. (3). Of note, the porosity in saturated porous media satisfies the following relationship (Biot 1941; Dvorkin & Nur 1993):

$$d\phi = \alpha d\varepsilon + dP/Q, \quad (22)$$

where ϕ denotes the porosity, ε denotes the strain and P denotes the pore pressure.

$$\frac{1}{Q} = \frac{\alpha - \phi}{K_0}, \quad \alpha = 1 - K_{\text{dry}}/K_0, \quad (23)$$

where K_{dry} is the dry bulk modulus and K_0 is the grain bulk modulus. The applied stress σ in the pore medium is related to ε and P , as follows:

$$d\sigma = K_{\text{dry}}d\varepsilon - \alpha dP. \quad (24)$$

The change in fluid mass in the pore space can be solved using eqs (22)–(24):

$$dm = \phi d\rho_f + \rho_f d\phi. \quad (25)$$

As the initial parameters (e.g. m_0 , P_0 , σ_0 , etc.) do not change with time, the mass flow from the pore can be obtained as follows:

$$\partial_t m = m_0 \left[\left(\frac{\phi}{K_f} + \frac{1}{Q} + \frac{\alpha^2}{K_{\text{dry}}} \right) \partial_t P + \frac{\alpha}{K_{\text{dry}}} \partial_t \sigma \right], \quad (26)$$

where $m_0 = \rho_{f0} V_0$ is the initial mass of the pore fluid. Substituting this expression into eq. (3) yields:

$$p_1 = \frac{1}{1 + i\omega\tau} (p_2 + i\omega\gamma\tau\sigma), \quad (27)$$

where

$$\tau = \eta V_0 A_1 \kappa \zeta$$

$$\gamma = B/A_1$$

$$A_i = \frac{\phi}{K_{fi}} + \frac{\alpha^2}{K_{\text{dry}}} + \frac{1}{Q_i}$$

$$B = -\frac{\alpha}{K_{\text{dry}}}. \quad (28)$$

As $V_0 \propto \tilde{a}^3$, and according to Chapman *et al.* (2002), $\zeta = \frac{\tilde{a}^2}{l}$ is the length scale controlling the fluid flow. And a and l are pore radius and squirt length as defined above. Therefore, the relaxation time $\tau \propto A_1 l \left(\frac{\eta}{\kappa} \right) \tilde{a}$ has a form similar to eq. (8) and Chapman (2009). Combining the fluid mass conservation eq. (10) and deriving it for the applied stress, we obtain the following:

$$\frac{dp_2}{d\sigma} = \frac{B - S_w A_1 S(\omega)}{S_w A_1 F(\omega) + (1 - S_w) A_2}, \quad (29)$$

where

$$S = \frac{i\omega\gamma\tau}{1 + i\omega\tau}$$

$$F = \frac{1}{1 + i\omega\tau}. \quad (30)$$

Combining eq. (27), eq. (29) can also be solved:

$$\frac{dp_1}{d\sigma} = F \frac{dp_2}{d\sigma} + S. \quad (31)$$

Then, the relationship between pore pressure and applied stress can be obtained. To solve the effective modulus, the method of Dvorkin *et al.* (1995) was used to describe the effective modulus for fluid-containing media.

$$K_{\text{eff}} = \frac{K_{\text{dry}}}{1 + \alpha dp/d\sigma}. \quad (32)$$

Volume averaging was used to approximate the effective bulk modulus for partial saturation. The Voigt and Reuss averages (Mavko *et al.* 2020) were as follows:

$$K_{\text{eff}1} = S_w \frac{K_{\text{dry}}}{1 + \alpha dP_1/d\sigma} + (1 - S_w) \frac{K_{\text{dry}}}{1 + \alpha dP_2/d\sigma} \quad (\text{Voigt average})$$

$$K_{\text{eff}2} = \frac{K_{\text{dry}}}{1 + \alpha S_w dP_1/d\sigma + \alpha (1 - S_w) dP_2/d\sigma} \quad (\text{Reuss average}). \quad (33)$$

The Voigt–Reuss–Hill (VRH) average (Mavko *et al.* 2020) could be used to calculate the the effective modulus for the general case:

$$K_{\text{eff}} = \frac{1}{2} \left(\frac{K_{\text{dry}}}{1 + \alpha S_w dP_1/d\sigma + \alpha (1 - S_w) dP_2/d\sigma} + \frac{S_w K_{\text{dry}}}{1 + \alpha dP_1/d\sigma} + \frac{(1 - S_w) K_{\text{dry}}}{1 + \alpha dP_2/d\sigma} \right). \quad (34)$$

Squirt-flow in the generalized pore system

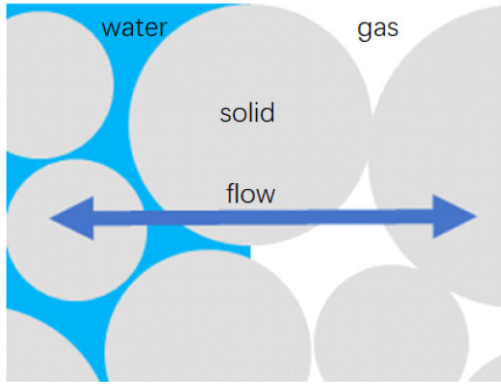


Figure 6. Schematic diagram of the generalized partial saturation model.

Eq. (34) provide the effective bulk modulus in a partially saturated medium. The new schematic diagram of the system is shown in Fig. 6. Comparing with the spherical pore assumption, the generalized model needs the dry modulus as input. However, it avoids the overestimation of the dry modulus compared to the spherical pore assumption. K_{dry} can be obtained by experimental tests or empirical formulas, such as Krief equation (Krief *et al.* 1990).

4.3 The characteristic frequency

To clarify the expression of characteristic frequency which is directly related to the effective modulus, eqs (29) and (31) are substituted into eq. (34) and organize it to obtain the following equation.

$$K_{\text{eff}} = K_{\text{dry}} \frac{M_{\varepsilon} + N_{\varepsilon}(i\omega\tau)}{M_{\sigma} + N_{\sigma}(i\omega\tau)}, \quad (35)$$

where

$$\begin{aligned} M_{\varepsilon} &= S_w A_1 + (1 - S_w) A_2 \\ N_{\varepsilon} &= (1 - S_w) A_2 \\ M_{\sigma} &= S_w A_1 + (1 - S_w) A_2 + \alpha S_w B + \alpha (1 - S_w) B \\ N_{\sigma} &= (1 - S_w) A_2 + \alpha (1 - S_w) B \\ &\quad + \alpha S_w (1 - S_w) A_2 \gamma - \alpha S_w (1 - S_w) A_1 \gamma. \end{aligned} \quad (36)$$

Carcione (2007) gave the stress–strain relation of the Zener model.

$$M(\omega) = M_R \left(\frac{1 + i\omega\tau_{\varepsilon}}{1 + i\omega\tau_{\sigma}} \right), \quad (37)$$

where M is the elasticity parameter, τ_{ε} and τ_{σ} are the relaxation time. And its characteristic frequency is

$$\begin{aligned} f_{\text{Zener}} &= \frac{1}{\tau_{\text{Zener}}} \\ \tau_{\text{Zener}} &= \sqrt{\tau_{\varepsilon} \tau_{\sigma}}. \end{aligned} \quad (38)$$

And we can write eq. (35) as

$$K_{\text{eff}} = K_{\text{dry}} \frac{M_{\varepsilon}}{M_{\sigma}} \frac{1 + \frac{N_{\varepsilon}}{M_{\varepsilon}}(i\omega\tau)}{1 + \frac{N_{\sigma}}{M_{\sigma}}(i\omega\tau)}. \quad (39)$$

Table 1. Parameters used for model comparison.

Parameter	Value
Grain bulk modulus, K_m (GPa)	38
Grain shear modulus, μ (GPa)	45
Grain density, ρ_s (kg m^{-3})	2560
Timescale (s)	3e-7
Bulk modulus of fluid 1 (GPa)	2.25
Bulk modulus of fluid 2 (GPa)	0.025
Viscosity of fluid 1 (cP)	1
Viscosity of fluid 2 (cP)	0.01
Density of fluid 1, ρ_{f1} (kg m^{-3})	1000
Density of fluid 2, ρ_{f2} (kg m^{-3})	1.26

Comparing eq. (39) with eq. (37), the characteristic frequency of the new model is

$$\begin{aligned} f_0 &= \frac{1}{\tau_0} \\ \tau_0 &= \tau \sqrt{\frac{N_{\varepsilon}}{M_{\varepsilon}} \frac{N_{\sigma}}{M_{\sigma}}}. \end{aligned} \quad (40)$$

4.4 Consistency with Gassmann equation

When $S_w = 1$ and angular frequency $\omega = 0$, the following can be obtained:

$$\begin{aligned} \frac{dp_1}{d\sigma} &= \frac{dp_2}{d\sigma} = \frac{B}{A_1} \\ K_{\text{sat}} &= K_{\text{eff}1} = K_{\text{eff}2} = \frac{K_{\text{dry}}}{1 + \alpha dp_1/d\sigma}. \end{aligned} \quad (41)$$

Bringing A1 and B into $dp/d\sigma$ yields:

$$\begin{aligned} \frac{dp_1}{d\sigma} &= -\left(\frac{\alpha}{K_{\text{dry}}}\right) / \left(\frac{\phi}{F} + \frac{\alpha^2}{K_{\text{dry}}}\right) \\ &= -\frac{1}{\alpha \left(1 + \frac{\phi K_{\text{dry}}}{\alpha^2 F}\right)}, \end{aligned} \quad (42)$$

where $1/F = 1/K_f + 1/(\phi Q)$. Of note, eq. (42) is the Skemp-ton coefficient. The Gassmann equation (Gassmann 1951) is obtained by substituting eq. (42) into the expression for K_{sat} in eq. (41).

$$K_{\text{eff}} = K_{\text{dry}} + \frac{(1 - K_{\text{dry}}/K_m)^2}{(1 - \phi)/K_m + \phi/K_f - K_{\text{dry}}/K_m^2}. \quad (43)$$

The above equation proves that the partial saturation model degenerates to the Gassmann equation when only one fluid is saturated.

5 APPLICATION TO THE DATA

The relaxation time τ can be estimated using the experimental data. Tillotson *et al.* (2014) and Amalokwu (2016) obtained relaxation times by fitting theoretical and experimental data. Here, we use the same method and fit the experimental results based on the $1/Q_P$ of samples. So that, we can obtain the relaxation time. The other input parameters are shown in Table 1 (Papageorgiou & Chapman 2015; Mavko *et al.* 2020). Additionally, the predictions of the new model and the White model are compared. And we can also get the patch size a_w in White model by fitting the predictions with the measured $1/Q_P$.

By fitting the experiment data, we can get the relaxation times $\tau = 2 \times 10^{-7}$ s, 2×10^{-7} s and 5×10^{-7} s for S4 (4.97 per cent),

S1 (21.03 per cent) and S2 (29.85 per cent), respectively. And we can also get the patch sizes $a_w = 3.5 \times 10^{-2}$ mm, 6×10^{-2} mm and 1.5×10^{-1} mm. Fig. 7 shows a comparison of the effective bulk moduli of the samples with the new model under partial saturation. Although we only fit the $1/Q_p$ of the samples. The predicted bulk modulus are generally consistent with the testing results. The predicted bulk modulus exhibited a slow increase at low-saturation conditions, whereas a faster increase in bulk modulus occurred at a high saturation level (near 80 per cent). Of note, the predicted bulk modulus for full saturation was significantly larger than the experimental results. The lower the porosity, the greater the difference between the predicted and experimental results. According to previous experiments (Verwer *et al.* 2010; Li *et al.* 2020), it is difficult to obtain 100 per cent saturation of the fluid owing to pore connectivity in tight rock. The rocks with low porosity usually have poorer pore connectivity. Therefore, the difference between the measured data and predicted results increases with decreasing porosity. As the Fig. 7 shows, the predictions of both models are consistent with the measured bulk modulus with water saturation. However, the new model predict the bulk modulus closer to the experimental data quantitatively.

Fig. 8 shows a comparison of the measured attenuation and predictions under different water saturation. The predicted results of both model for the attenuation fit the measured data. Both models predict the maximum attenuation higher than the water saturation of 50 per cent, in agreement with the results on sandstone reported by Murphy III (1984). However, the predicted attenuations are 0 at the saturation of 0 and 100 per cent, which is different from the test results. The reason is the two models only aim at the partial saturation induced attenuation. So, they does not include reasons such as pore structure inhomogeneity. The details will be discussed later.

6 DISCUSSION

In this discussion, the parameters of the 4.97 per cent porosity sample were used to illustrate the characteristics of the new model.

6.1 Influence of frequency

The characteristic frequency f_0 is mainly related to the relaxation time τ in the new model (eq. 40), whereas in the White model it is mainly related to the radius of the patch a_w . To ensure our predictions fit the $1/Q_p$ of S4 (4.97 per cent), we obtained $\tau = 2 \times 10^{-7}$ s and $a_w = 2.3 \times 10^{-2}$ mm. However, the resulting f_0 (Fig. 9) is high and does not match the available results of low frequency tests (e.g. Tisato & Quintal 2013; Spencer & Shine 2016). There are several possible reasons for this. First, partial saturation is only one of the causes of dispersion, and f_0 obtained by fitting the predictions to experimental data. So, the predicted f_0 may differ from the attenuation induced by partial saturation. Secondly, low-frequency test results may not be sensitive to inhomogeneous distribution of fluids with small scales.

Fig. 10 shows the frequency dependent $dp_1/d\sigma$ and $dp_2/d\sigma$, where dp_1 is pore pressure for water-saturated pores and dp_2 is pore pressure for gas-saturated pores, σ is the applied stress. When the frequency is low, the fluid pressures in the different pores are in equilibrium. As the frequency increases, $dp_1/d\sigma$ gradually tends to equals the Skempton coefficient (Skempton 1954) of the fully water-saturation, while $dp_2/d\sigma$ gradually tends to equals the Skempton

coefficient of the fully gas-saturation. This indicating that at high frequency, the water-saturated pore and gas-saturated pore are both isolated, and the fluids could not flow between the two types of pores. Compared to low frequency, the rock has a higher elastic modulus in this case. This is consist to the description of Dvorkin *et al.* (1995) for the fluid motion in rocks under low- and high-frequency. Papageorgiou & Chapman (2017) introduced a constant q to calculate the pore pressure relationship between WP and GP and considered the value of q to be vary from $K_{\text{gas}}/K_{\text{water}}$ (the ratio of gas and water bulk modulus) to 1. Our results suggest that this value is frequency-dependent. Moreover, the frequency-dependent pore pressure should also be considered in the fracture-pore medium model with partial saturation.

Fig. 11 shows the variation of bulk modulus with saturation at different frequencies. The low-frequency limit of the new model is consistent with the White model, whereas the high-frequency limit is slightly different from that of the White model. Because the modulus of both the new model and the White model can be obtained by Gassmann's relation with an effective fluid modulus equal to the Reuss average of the fluid moduli in the low frequency limit. However, in the high frequency limit, the new model satisfies the VRH average, whereas White model is not. As for the VRH average can be useful for moduli estimation (Mavko *et al.* 2020) and the result differs very little from the White model, we think it is reasonable.

6.2 Influence of fluids

Fig. 12 shows the predicted results of bulk modulus and attenuation when saturated with various multiphase fluids in the pore space. The results of mixing different liquids such as water, gasoline and heavy oil with gas were analysed, with a gas saturation of 20 per cent. For kerosene, the bulk modulus is 1.27 GPa and the viscosity is 2.1 cP (Ba *et al.* 2019). The bulk modulus of heavy oil is 3.45 GPa and the viscosity is 1.75×10^5 cP (Zhao *et al.* 2017). Fig. 12(a) shows the characteristic frequency decrease with increasing fluid viscosity. Due to the large difference between the bulk moduli of liquid and gas, this difference in fluid compressibility leads to the increase of pore pressure gradient under the applied pressure. This induce strong fluid motion between different regions, resulting in stronger attenuation. At low-frequency, the difference of the effective bulk modulus caused by various multiphase fluid saturation is small. This indicates that when the frequency approaches 0, it is effective to calculate the effect of fluid on the effective bulk modulus through Wood averaging. It should be noted that the prediction results of the Wood equation are greatly influenced by fluids with small bulk moduli. At high frequencies, the effective modulus is more influenced by the liquid with high bulk moduli than that at low frequency. These are consistent with the experimental results reported by Spencer & Shine (2016). Notably, the attenuation peak is close to the frequency of laboratory ultrasonic measurements when water and gas are saturated in the pore space. The new model describes the pore-to-pore fluid motion, while the pores in synthetic sandstone samples are mainly in grain scale (i.e. pores with a diameter less than 1/10 of the wavelength). Therefore, the fluid motion discussed in Fig. 12 can be considered in grain scale, which also cause a higher characteristic frequency.

It should be noted that the local flow is not the sole mechanism to investigate the influence of multi-phase fluid on rock properties. Various studies have explored different aspects of this phenomenon. For example, Cadoret (1995) demonstrated the elastic weakening of very-dry rock upon moistening, and attributed the velocity drop

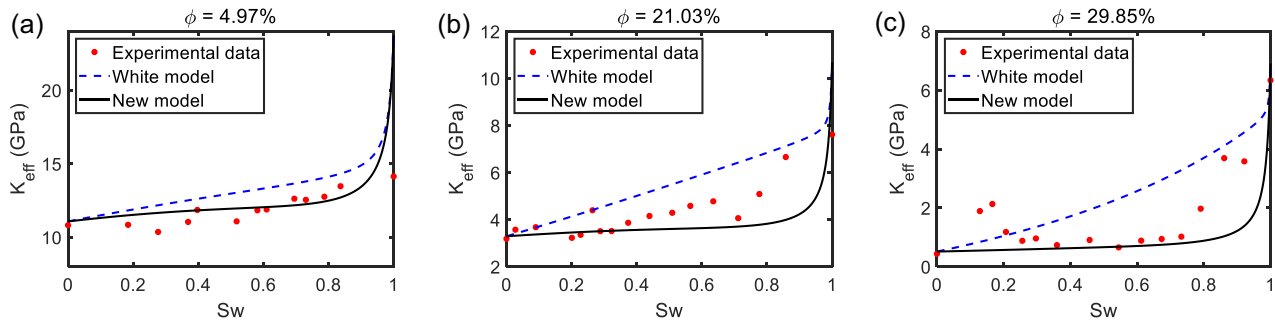


Figure 7. Comparison of the measured bulk moduli and predictions of the new model for samples with porosities of (a) 4.97 per cent, (b) 21.03 per cent and (c) 29.85 per cent.

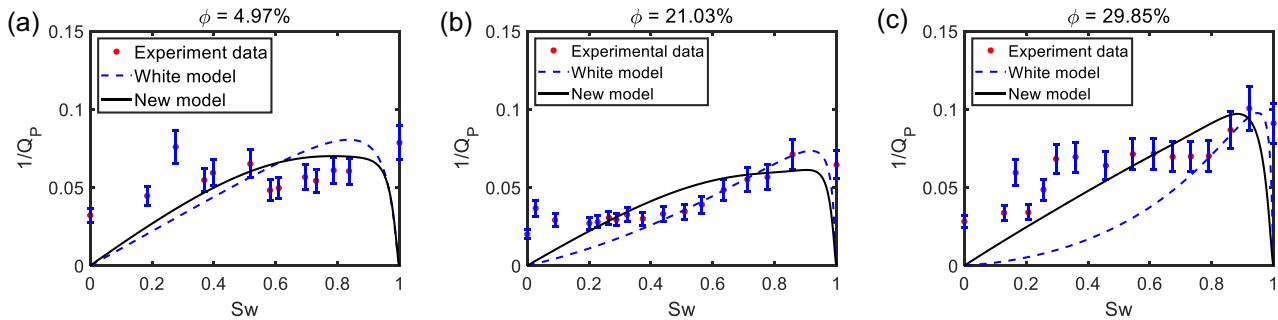


Figure 8. Comparison of measured attenuations and predictions of the new model for samples with porosities of (a) 4.97 per cent, (b) 21.03 per cent and (c) 29.85 per cent.

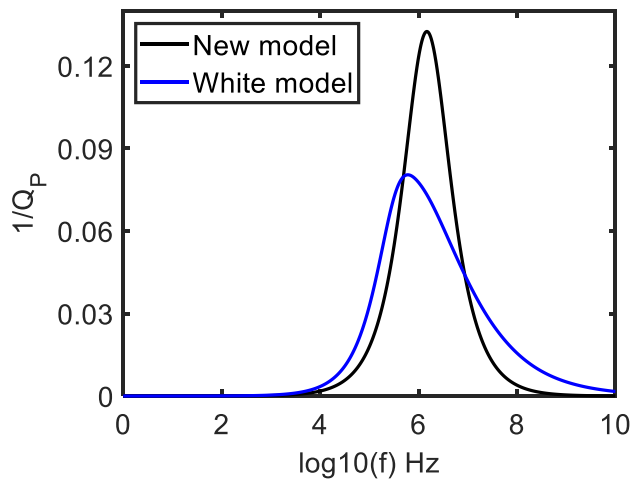


Figure 9. The predicted attenuation versus frequency curves for sample S4 with 80 per cent water saturation.

to the softening of cements. In the experiments of Pimienta *et al.* (2014), the microstructure (grain contact) is considered to be a main factor contributing to the different evolution in P -wave velocity with relative humidity between sandstone and limestone at a low water saturation. Additionally, Pimienta *et al.* (2019) carried out a comparative experiment demonstrating the velocity and attenuation during the spontaneous water imbibition and moisture adsorption. They suggested that two competing mechanisms, moisture adsorption and elastic stiffening of water, will influence the elastic modulus. These mechanisms are important to investigate the low water saturation characteristics of the rock. Another noteworthy observation by Lebedev *et al.* (2009) was a sharp increase in wave

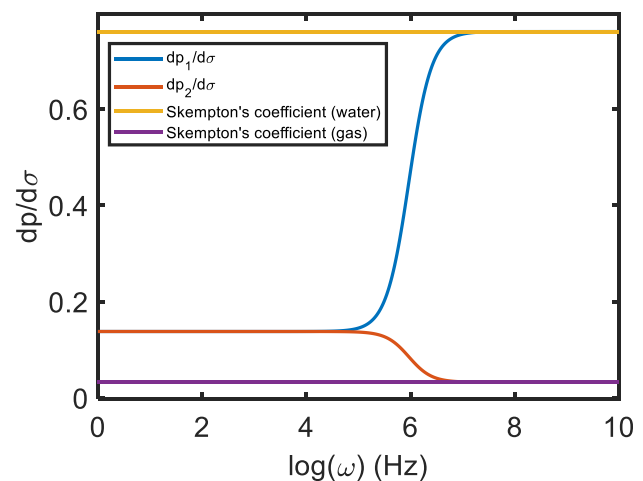


Figure 10. Pore pressures in the new model are in equilibrium at low frequencies, while the ratios of different pore pressures and applied stresses at high frequencies are close to the corresponding Skempton coefficients.

velocity at an intermediate water saturation (around 40 per cent). They claimed the phenomena attributed to the interplay between Biot's slow wave and the local flow. Furthermore, Ma & Ba (2020) conducted experiments that revealed stronger coda attenuation under partial saturation, indicating the importance of scattering as a mechanism for inducing wave attenuation in partially saturated rocks. However, in our current study, these mechanisms were not taken into consideration. And addressing these mechanisms will be a focus of our future work.

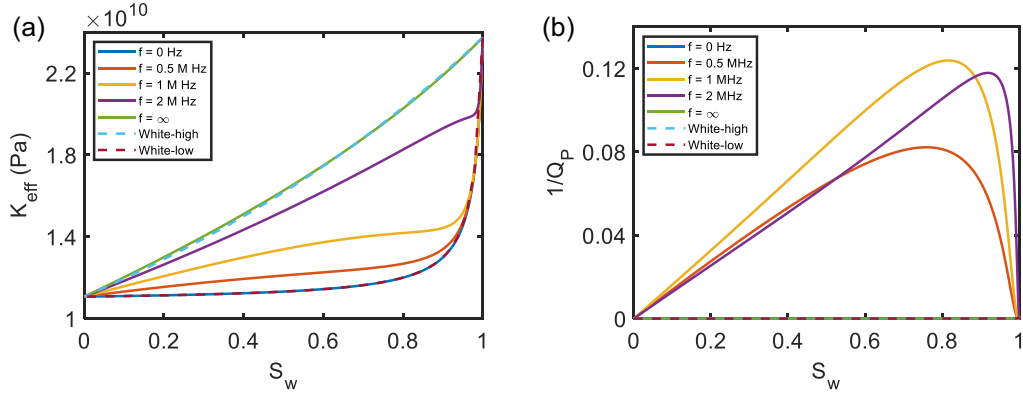


Figure 11. The variation of (a) bulk modulus and (b) attenuation with water saturation for a partially saturated sand for different relative frequencies.

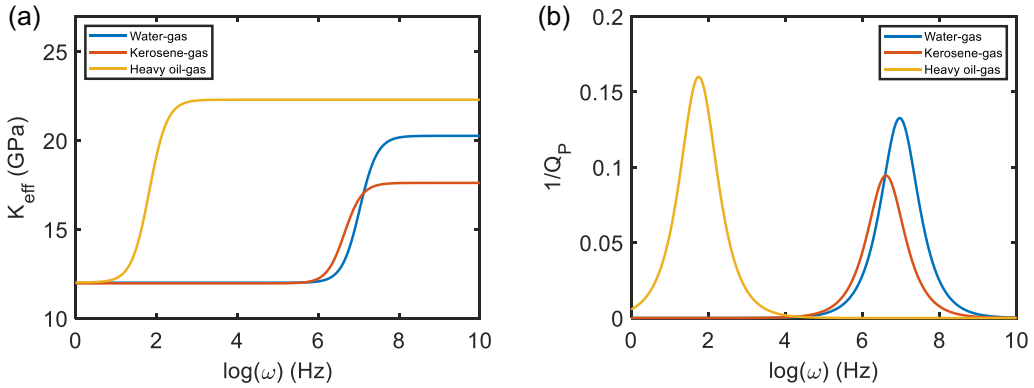


Figure 12. Variation of (a) bulk modulus and (b) attenuation with frequency for partially saturated sand imbibed with different fluids.

6.3 Influence of inhomogeneity of pore structure

As the Figs 8 and 11 show, the attenuation is not 0 at the water saturation of 0 and 100 per cent. However, it conflicts both the new model and White model's predictions. The reason is partial saturation is one of the reasons to induce local flow. Murphy III (1984) identified three main factors that contribute to attenuation in partially saturated rocks: gas detection, contact relaxation and frame modulus dispersion. The latter two can be attributed to attenuation caused by the pore structure. Pimienta *et al.* (2015a) conducted experiments that showed a significant decrease in bulk modulus attenuation of saturated rocks as the effective pressure increased. It is due to the effective pressure causing soft pores to close or decrease in volume, resulting in less difference in pore compressibility and finally leading to a decrease in fluid flow. These highlight the importance of pore structure heterogeneity in fluid-bearing rocks as a cause of dispersion and attenuation.

To investigate the impact of pore structure inhomogeneity in this experiment, we tested the dry samples under different confining pressure using the system BenchLab 7000EX. We set E_h to the Young's modulus of the rock at a certain pressure and E is the initial Young's modulus. There are at least two reasons for the change in E during pressurization: the crack closure, and pore deformation or volume reduction due to pressure even in the absence of cracks. We know soft pores are more easily deformed by pressure than stiff ones, causing an increase in elastic modulus when more soft pores are subjected to the same pressure. So, the E_h/E can indicate the percentage of soft pores, such as cracks. Fig. 13 shows the variation of E_h/E with pressure.

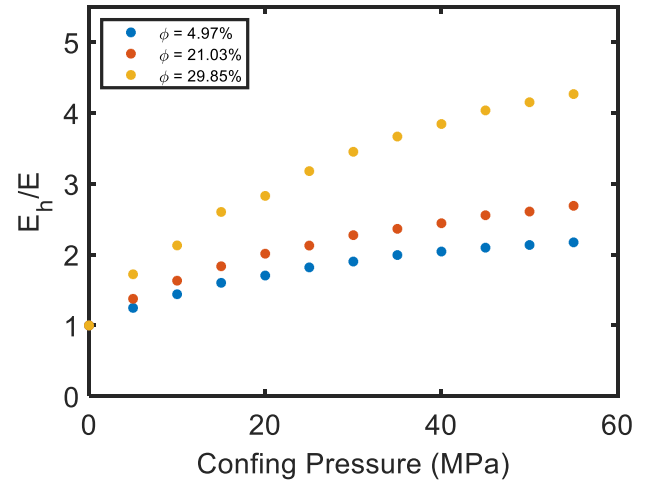


Figure 13. Variation of E_h/E with confining pressure. Under the same pressure, a higher E_h/E indicates more soft pores.

At the pressure of 55 MPa, the values of E_h/E for the samples S4 (4.79 per cent), S1 (21.03 per cent) and S2 (29.85 per cent) are 2.17, 2.69 and 4.26, respectively. We can observe that S2 has a significant higher E_h/E . Since 29.85 per cent is higher than the consolidated porosity, the sample likely contains more soft pores such as incomplete contact of grains, which is consistent with the calculated results. The new model does not include the fluid flow

induced by pore structure inhomogeneity. However, the compressibility of the pore still has impact on the new model. For example, the compressibility of pores is related to the K_{dry} of the rock. The dispersion and attenuation will be changed by various K_{dry} . And the eq. (40) indicate that the characteristic frequency of the new model is also related to the rock frame (K_{dry} and ϕ). And the combination of fluid distribution and pore structure requires the future study.

7 CONCLUSION

A partial saturation model based on the Chapman model was developed by discussing the squirt-flow between pores containing different fluids. The new model with spherical pore assumption will overestimates the dry bulk modulus and can lead to inaccurate predictions of dispersion and attenuation. Therefore, we combined pore medium theory with this idea and proposed a generalized partially saturated squirt model. By comparing the experimental data with the predictions and analysing the model, the following conclusions were drawn.

The new model can provide reasonable predictions of bulk modulus and P -wave attenuation under partial saturation. When the attenuations fit the experimental data, the new model's bulk modulus predication matches the measured results well.

The new model is fully consistent with the Gassmann equation when saturated with a single fluid. The pore pressure varies with frequency in the new model, which differs from the assumption of Papageorgiou & Chapman (2017). This indicating that the pressure of the pores containing different fluids is balanced at low frequencies. At high frequencies, the pores are relatively closed which agree with the results of Dvorkin *et al.* (1995). Hence, the high- and low-frequency limits of the new model are consistent with White's model. The predictions of the new model with different fluids are similar to the previous laboratory study reported by Spencer & Shine (2016).

Based on the experimental calibration and numerical analyses, the new model could be a useful extension of Chapman's model. In addition, the new model can provide reliable results at predicting the bulk modulus and the attenuation of rocks with homogeneous and well-connected pore structures. However, the pore structure inhomogeneity must be considered in future work.

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DATA AVAILABILITY

Data associated with this research are available and can be obtained by contacting the corresponding author.

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APPENDIX: MEASURED VELOCITY AND $1/Q_P$ **Table A1.** Partially saturated experiment results for S-1.

Water saturation (per cent)	Sample S-1			
	ρ (kg m ⁻³)	P -wave velocity (m s ⁻¹)	S -wave velocity (m s ⁻¹)	$1/Q_P$
100	2142	2828	1826	0.064452
86	2112	2742	1810	0.071132
78	2095	2607	1810	0.056553
71	2081	2514	1810	0.055115
64	2065	2575	1800	0.048426
56	2050	2565	1805	0.039009
51	2038	2544	1810	0.034534
44	2024	2534	1810	0.033042
37	2010	2554	1858	0.029992
32	1999	2554	1891	0.032471
29	1992	2544	1880	0.029425
26	1987	2575	1820	0.030581
23	1979	2565	1914	0.027968
20	1973	2565	1926	0.027091
9	1950	2596	1908	0.029102
3	1937	2629	1949	0.036695
0	1931	2629	1986	0.020058

Table A2. Partially saturated experiment results for S-2.

Water saturation (per cent)	Sample S-2			
	ρ (kg m ⁻³)	P -wave velocity (m s ⁻¹)	S -wave velocity (m s ⁻¹)	$1/Q_P$
100	2008	2100	970	0.090957
92	1985	1716	925	0.10056
86	1967	1698	869	0.086791
79	1946	1399	841	0.070039
73	1929	1210	838	0.069821
67	1911	1202	845	0.069567
61	1893	1197	851	0.07164
54	1873	1156	860	0.071323
46	1847	1224	869	0.064005
36	1817	1208	890	0.069453
30	1798	1258	888	0.068227
26	1786	1273	919	0.048512
21	1772	1356	938	0.03409
17	1760	1577	979	0.059434
13	1748	1600	1053	0.033792
0	1710	1503	1226	0.028258

Table A3. Partially saturated experiment results for S-4.

Water saturation (per cent)	Sample S-4			
	ρ (kg m ⁻³)	P -wave velocity (m s ⁻¹)	S -wave velocity (m s ⁻¹)	$1/Q_P$
100	2202	3617	2271	0.078635
84	2181	3549	2235	0.060413
79	2163	3505	2235	0.060976
73	2145	3475	2215	0.054267
70	2129	3483	2218	0.056634
61	2119	3435	2215	0.049674
58	2114	3429	2211	0.048175
52	2108	3301	2119	0.06508
40	2102	3462	2244	0.05937
37	2098	3301	2119	0.054652
28	2098	3172	2021	0.075924
18	2094	3400	2244	0.044617
0	2090	3441	2262	0.03219